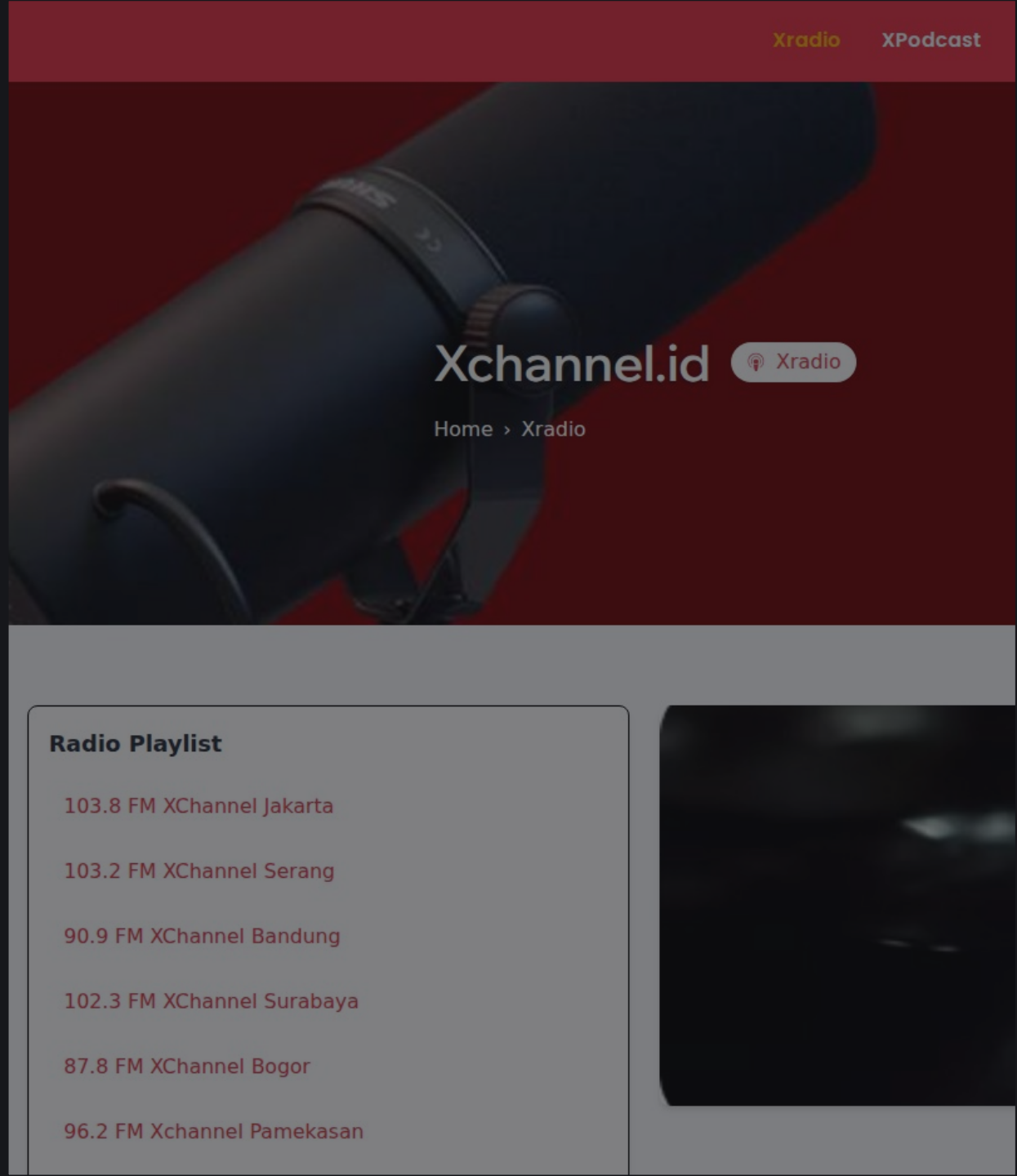
XRADIO · DASHBOARD

XChannel

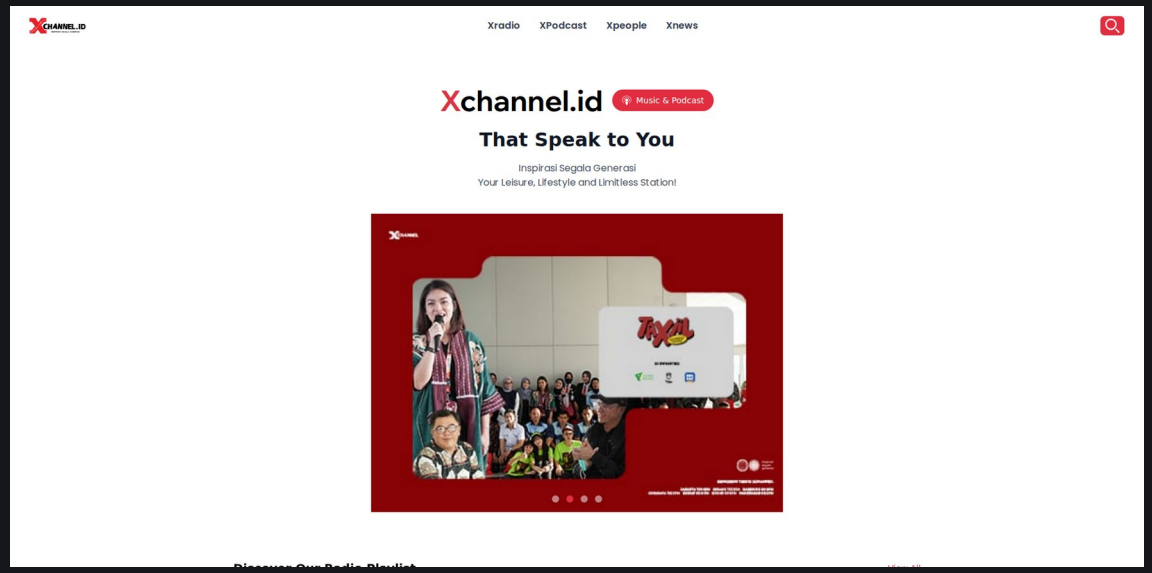
The broadcast schedule for local radio in several cities, and the system behind it

Enrico Saputra · Backend developer, Laravel dashboard

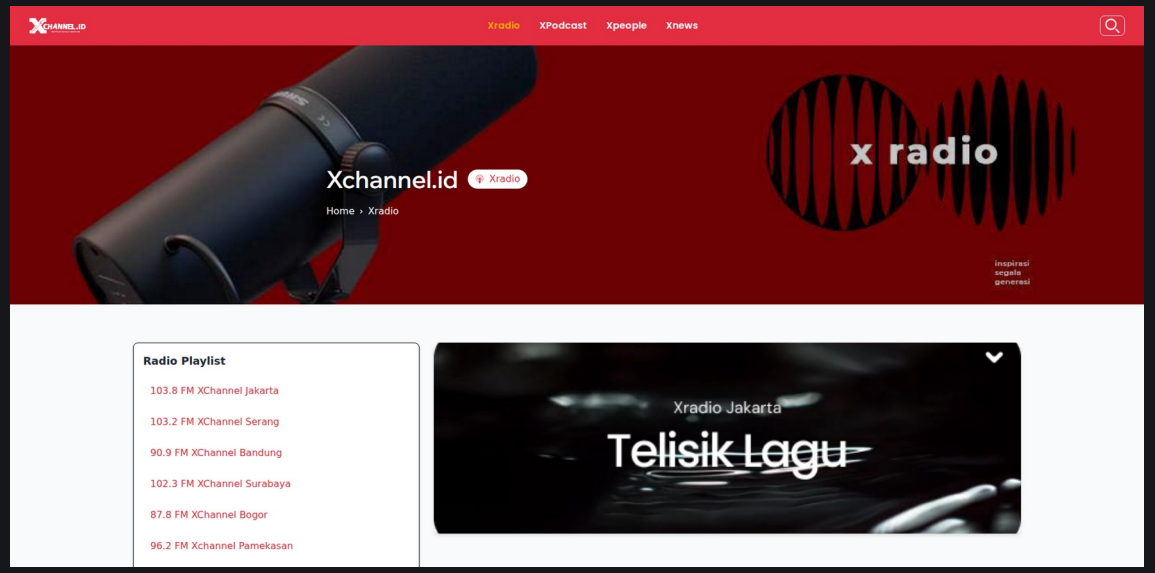
Client code and data are private. The demo rebuilds the radio part of the dashboard on the real data model, with made-up data.



xchannel.id, the side listeners see



Home: radio, podcasts, people and news



Xradio: stations per city and the show that is playing

The stations, shows, hosts and schedule on the Xradio page are managed in the Laravel dashboard. That is the part this demo rebuilds.

Screenshots of xchannel.id, October 2026

What I built at XChannel

The XChannel app plays each station's stream and shows what is on air. The broadcast team runs all of it from the dashboard.

What I built

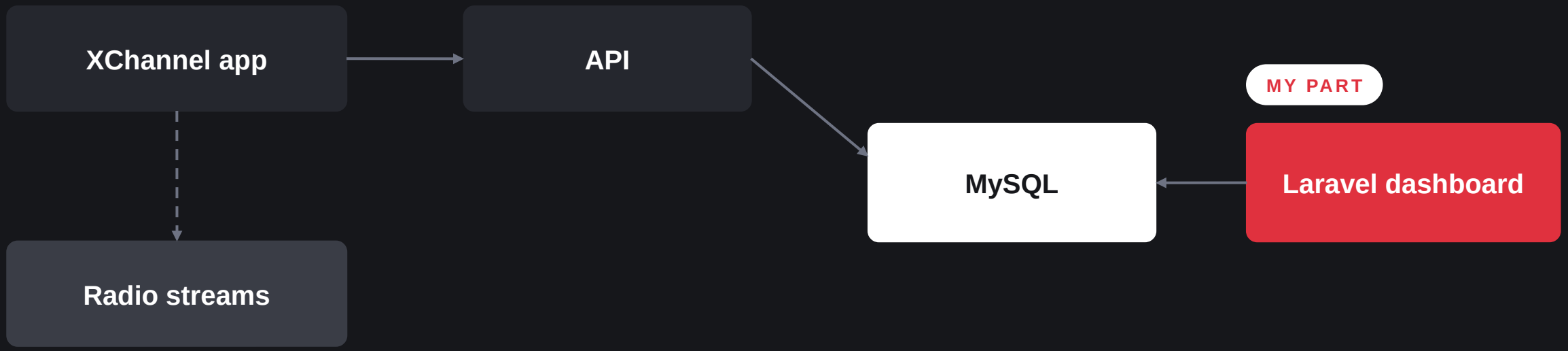
- Stations: city, frequency, stream URL and status
- Shows and hosts
- Each station's weekly schedule, which the app reads
- Podcasts, chart, video, news: same dashboard, outside this demo

8
stations on xchannel.id today

7 → 1
per-city schedule tables become one

24/7
empty hours play automated music

One database, two readers



The dashboard writes

Stations, shows, hosts and the schedule.

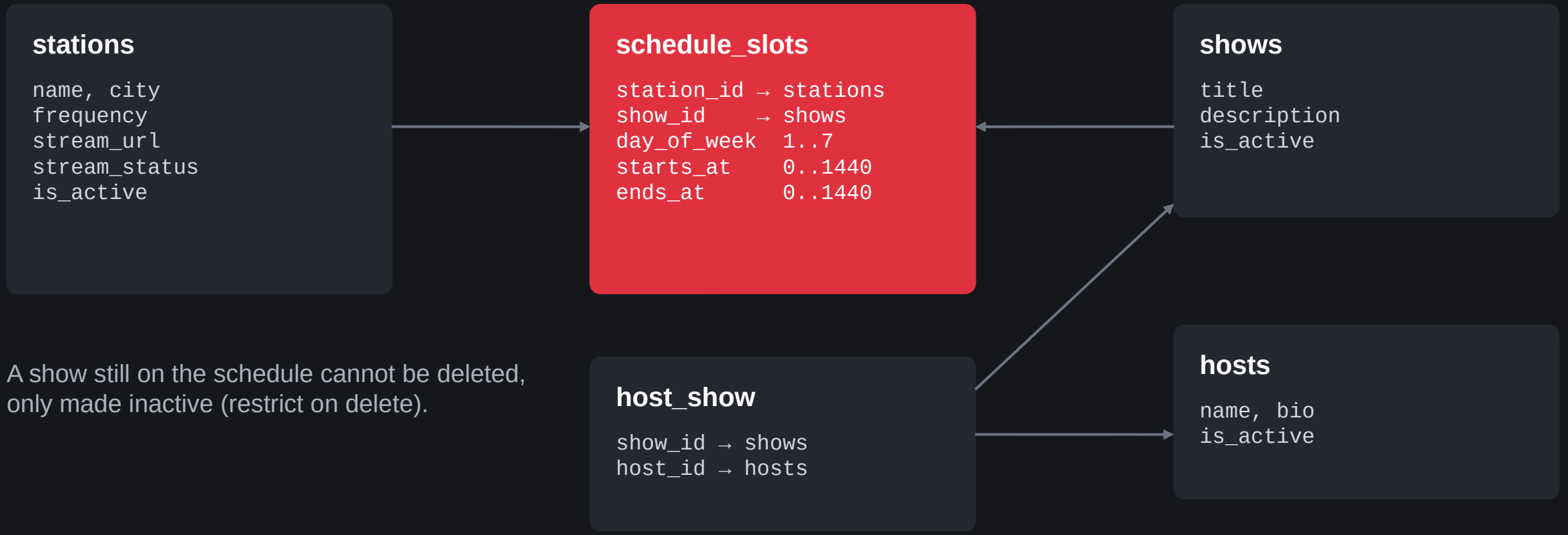
The app reads

"What is on air now?" through its API.

Rules on save

An overlapping schedule never reaches a listener.

Four tables and one pivot



A show still on the schedule cannot be deleted, only made inactive (restrict on delete).

From a table per city to one table

Production

7 TABLES

```
radio.api_jadwal = 'jadwal_radio_woman'

jdwl_radio_bandung
jdwl_radio_surabaya
jdwl_radio_suramadu  ...
  judul_siaran, hari_siaran,
  waktu_siaran, penyiar (text)
```

A new station means a new table. One name in api_jadwal does not match the real table.

Demo

1 TABLE

```
schedule_slots
  station_id  -> stations
  show_id     -> shows
  day_of_week, starts_at, ends_at

host_show (show_id, host_id)
```

A new station is one row. Foreign keys keep it consistent.

No-downtime migration: fill the new table from the old ones, keep a view per city for the old API, move readers one by one.

Time as numbers, not text

In production

`senin-jumat`	`Senin- Jumat`
`Monday-Friday`	`6:00 AM - 10:00 AM`
`08.00-09.00`	`20:00-24:00`

In the demo

450

minutes = 07:30

1440

minutes = 24:00

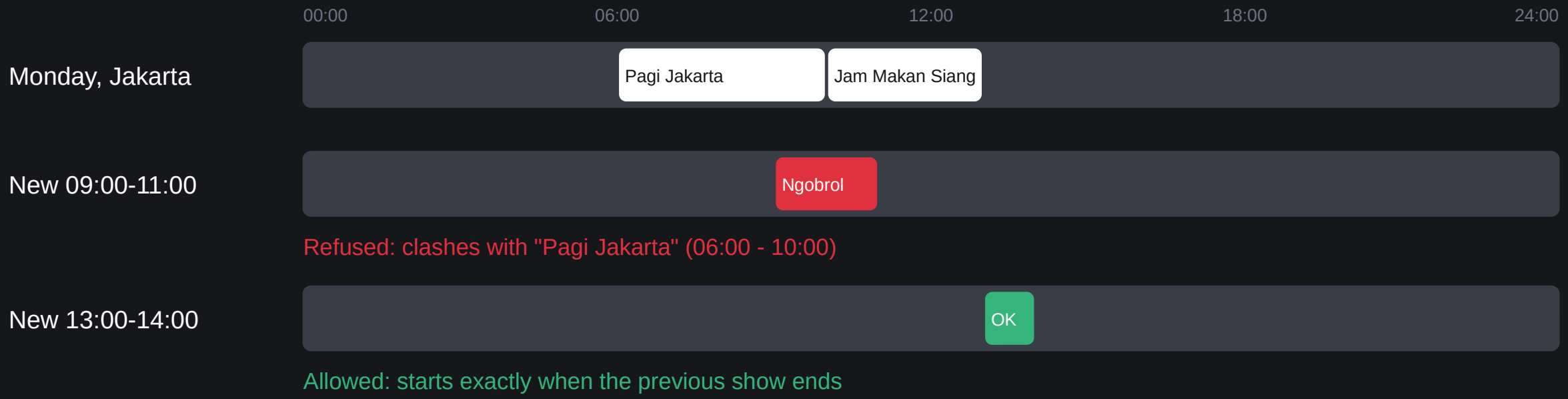
day_of_week 1-7 (ISO), one row per day. The form takes HH:MM only.

Two slots overlap when each starts before the other ends

```
starts_at < :end  AND  ends_at > :start
```

Touching (10:00 then 10:00) is fine.

The overlap rule on save



A clear message

Day, time and the show in the way.

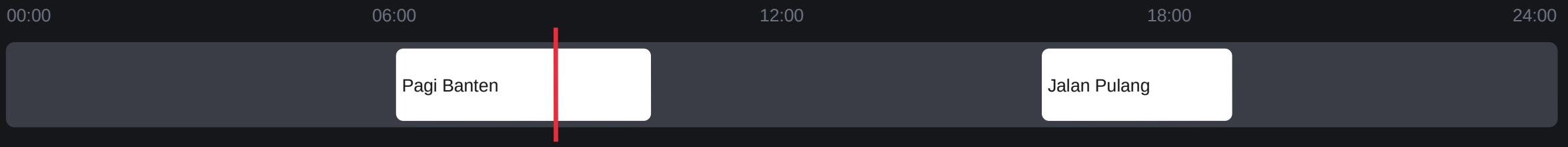
All days or none

Monday to Friday in one transaction.

Past midnight

Two slots: until 24:00, then from 00:00.

On air, up next, and empty hours



XChannel Banten, Monday 08:30. Red line = now; grey = automated music.

On air

Today's slot with start <= now < end. One indexed query.

Up next

The first slot after now, rolling into the next days for up to a week.

Empty hours

Worked out from the schedule, not stored. Airtime per station out of 168 hours.

The trade-offs I chose

Decision	Why	What it costs
Minutes since midnight, not a TIME column	24:00 is valid; overlap checks are integer compares	Every screen has to convert (Clock::format)
Late shows are split at midnight	One slot = one day, so every query stays simple	Friday's late show becomes two rows
Co-hosts count the full slot	Both of them work the whole slot	Per-host totals add up to more than per-station totals
Deactivate, not delete	Schedule history stays intact	Lists need an active filter

Why this stack, and what else I considered

Choice	Why	What I considered instead
Laravel 10	Same as production; forms and validation are quick	An admin generator (Filament): faster, but the schedule rules end up scattered in config
MySQL	The production database	PostgreSQL has exclusion constraints that could block overlaps in the database. Not used, to match production; the rule lives in a service
Blade + Tailwind	The timelines are plain HTML and CSS	Vue or React: not needed for these admin pages
PHPUnit	19 feature tests on the schedule rules	Pest: a matter of taste; PHPUnit ships with Laravel 10

The overlap check is one query

```
public function clash(int $stationId, Day $day,
    int $start, int $end, ?int $ignoreId = null)
{
    // Two ranges overlap when each one starts
    // before the other ends. Touching is fine.
    return ScheduleSlot::with('show')
        ->where('station_id', $stationId)
        ->where('day_of_week', $day->value)
        ->where('starts_at', '<', $end)
        ->where('ends_at', '>', $start)
        ->when($ignoreId, fn ($q) =>
            $q->whereKeyNot($ignoreId))
        ->first();
}
```

Rules in one service

Clash, schedule, move, on air, up next, gaps.

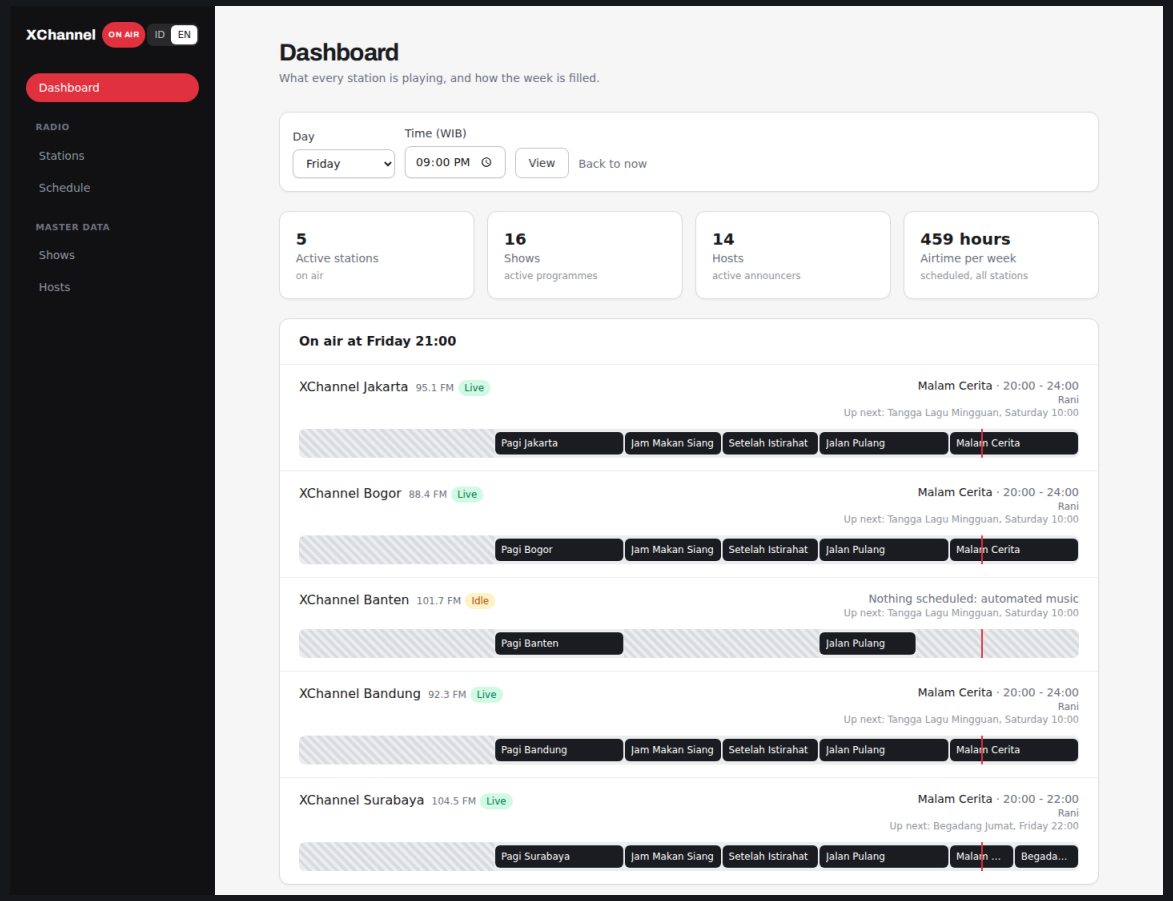
Enums, not numbers

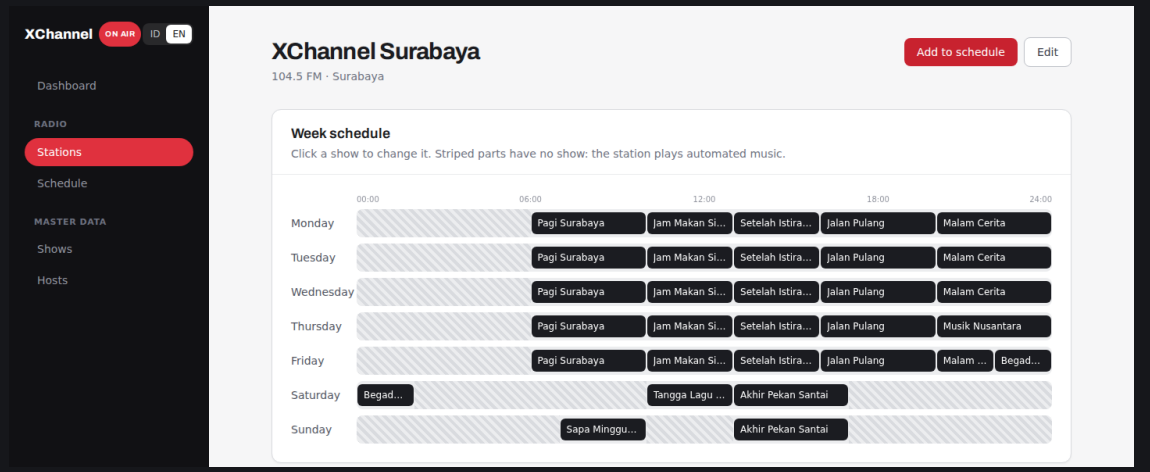
Day (ISO 1-7), StreamStatus.

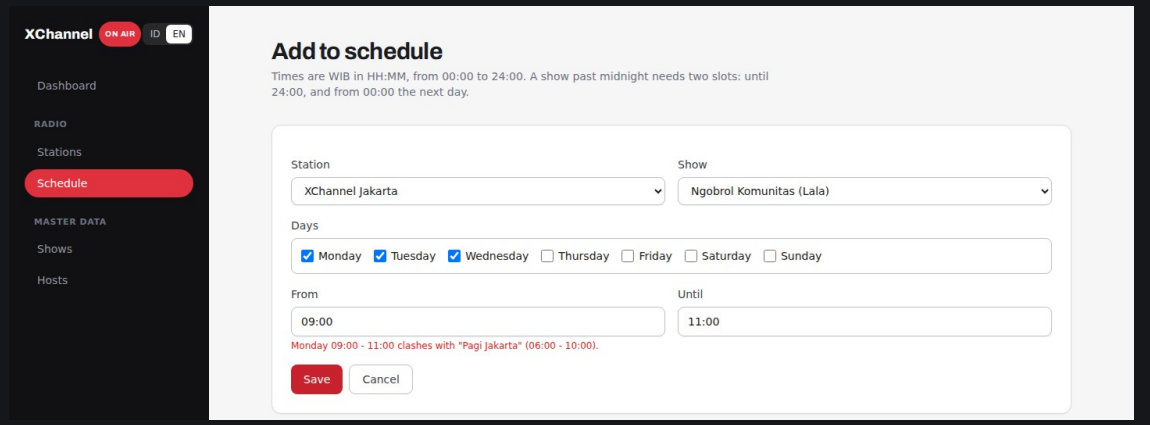
Thin controllers

FormRequests validate, Clock parses times.

Demo: dashboard, schedule, a refused slot







Dashboard: on air and airtime

What is not done yet

Not done yet	First step
Production still has a table per city	Fill schedule_slots from the old tables, keep a view per city for the old API, then move readers over gradually
Every time is assumed to be WIB	A time zone column on stations, for stations in WITA or WIT
No one-off changes yet	A per-date override table (holidays, specials) on top of the weekly pattern
Stream status is set by hand	A stream health check that sets live / idle / offline

Demo, slides and the real product

Live demo	https://xchannel-dashboard-demo.vercel.app
Slides (PDF)	https://xchannel-dashboard-demo.vercel.app/xchannel-walkthrough.pdf
Slides (PPTX)	https://xchannel-dashboard-demo.vercel.app/xchannel-walkthrough.pptx
Real product	https://xchannel.id/radio

The demo code is in a private GitHub repo; access is given during the interview. The dashboard speaks Bahasa Indonesia and English, and its data is restored every 3 hours.